

1) Solve the following matrix multiplication problems

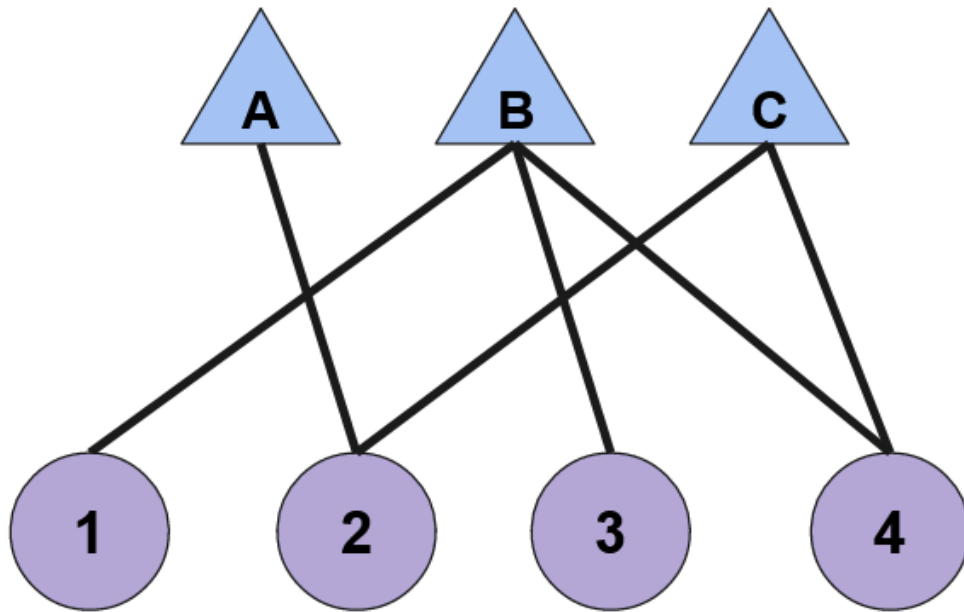
$$\begin{bmatrix} 8 & 9 & 3 \\ 3 & 2 & 2 \end{bmatrix} * \begin{bmatrix} 5 & 4 \\ 1 & 2 \\ 3 & 2 \end{bmatrix} =$$

$$\begin{bmatrix} 8 \\ 2 \end{bmatrix} * \begin{bmatrix} 8 & 5 \\ 7 & 6 \\ 9 & 1 \end{bmatrix} =$$

$$\begin{bmatrix} 2 & 1 & 4 \end{bmatrix} * \begin{bmatrix} 3 \\ 3 \\ 0 \end{bmatrix} =$$

$$\begin{bmatrix} 5 & 5 \\ 3 & 2 \\ 3 & 1 \end{bmatrix} * \begin{bmatrix} 3 \\ 2 \end{bmatrix} =$$

2) *Affiliation Network*



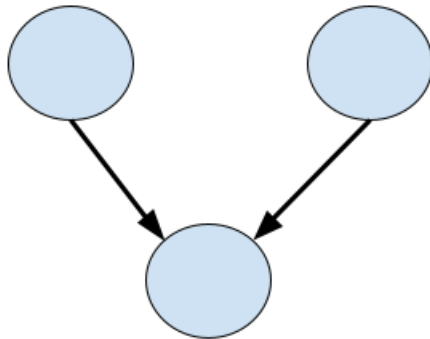
a) What is the affiliation matrix of the above graph?

b) What is the member overlap matrix for the above affiliation matrix?

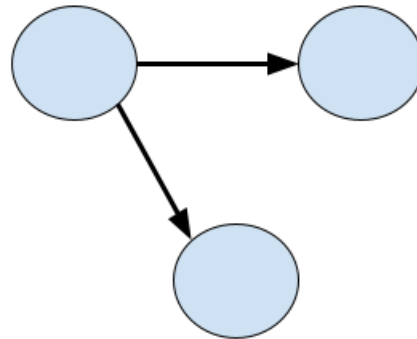
3) What do you do according to the theory?

- a. Tom is a rude person. According to karma theory you?
- b. Everyone likes Aruna. According to status theory you?
- c. No one likes Christophe. According to solidarity theory you?
- d. Manuel is a friendly person who's nice to all. According to karma theory you?

4) The following two triads are hierarchies. First, do winner, loser, and/or by-stander effects create the dominance order for (a) and (b)? Close the triad by making the hierarchy transitive. Justify the direction of the transitivity. Is the example of an attribute or interaction-based model?

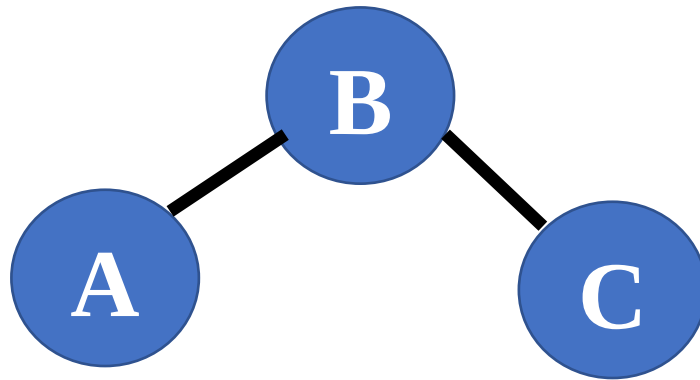


(a)



(b)

5. Multiply the adjacency matrix by itself. How many walks of length 2 start or end with A?



6. Definitions

Two-Mode

Affiliation Network- People to groups

Bipartite Graph

Affiliation Matrix

Matrix Transpose

Co-Membership Matrix

Group Overlap Matrix